5.8 A Hidden Distance in the Disc

WORKSHEET

The question

STEP III 2013, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s3-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself How does $|z-w| \leq |z|+|w|$ follow from a diagram? (Think of z and $-w$ as vectors.)

STAGE 1 Expanding the moduli

Covers: Part (i)

Ask yourself Use $|a|^2 = aa^*$. Expand $|z-w|^2 = (z-w)(z^*-w^*)$ and compare with $(|z|+|w|)^2$.

Pause and try Why is E real? For $E \geq 0$, use the triangle inequality.

STAGE 2 The second identity

Covers: Part (ii)

Ask yourself Expand $|1 - zw^*|^2$ in the same way.

STAGE 3 The inequality

Covers: the final inequality, and the case $|z| < 1$, $|w| < 1$

Ask yourself Write $A = (|z| + |w|)^2$ and $B = (1 + |zw|)^2$. By parts (i) and (ii), the inequality squared says $\frac{A-E}{B-E} \leq \frac{A}{B}$. When is that true?

Pause and try Factorise $1 + |z||w| - |z| - |w|$. When is it non-negative? Does that also cover $|z| < 1$ and $|w| < 1$?

What's really going on

Non-Euclidean geometry

For two thousand years mathematicians tried to prove Euclid's fifth postulate (through a point not on a line, there is exactly one parallel to the line) from his other axioms. In the 1820s and 1830s Nikolai Lobachevsky, János Bolyai and (privately) Gauss realised that it cannot be done: there is a perfectly consistent geometry in which there are infinitely many parallels. In 1882 Henri Poincaré found a beautiful model of it inside the unit disc. Points are points of the open disc; "straight lines" are diameters and circular arcs that meet the boundary circle at right angles; and the distance between z and w is

$$d(z,w) = 2\tanh^{-1}\rho(z,w), \qquad \text{where} \qquad \rho(z,w) = \frac{|z-w|}{|1-zw^*|}.$$

As a point approaches the boundary circle, its distance from the centre tends to infinity: the disc is a whole infinite plane, seen through a distorting lens.

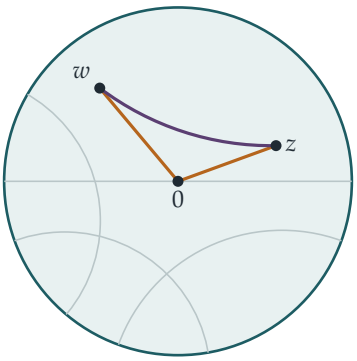


Figure 5.8.1 The Poincaré disc. Hyperbolic "straight lines" are diameters or arcs of circles meeting the boundary at right angles (grey). The hyperbolic triangle with vertices 0, z and w has two sides along diameters (amber) and one along an arc (plum). The inequality in the question says that the plum side is no longer, in hyperbolic distance, than the other two put together.

The triangle inequality in disguise

Here is what the question proved, for $|z|, |w| < 1$. Write $|z| = \tanh a$ and $|w| = \tanh b$, so that the hyperbolic distances from 0 to z and w are $2a$ and $2b$. Then by the addition formula for tanh from Section 3.8,

$$\frac{|z|+|w|}{1+|zw|} = \frac{\tanh a + \tanh b}{1+\tanh a\tanh b} = \tanh(a+b).$$

So the inequality says $\tanh\left(\frac{1}{2}d(z,w)\right) \le \tanh(a+b)$, that is,

$$d(z,w) \le 2a + 2b = d(z,0) + d(0,w).$$

This is the triangle inequality for hyperbolic distance (Figure 5.8.1). The relativistic velocity-addition formula and hyperbolic geometry are secretly the same thing: the

set of possible velocities in special relativity, with the right notion of distance, *is* a hyperbolic space. We return to this in Section 7.6.

Symmetries of the disc

Why is ρ the right quantity? The maps $z \mapsto \dfrac{z - a}{1 - a^{*}z}$, for $|a| < 1$, send the unit disc to itself, preserve the hyperbolic distance, and move a to the centre. They are the hyperbolic analogues of translations, and $\rho(z, w)$ is exactly |image of z| when w is moved to the centre. These *Möbius transformations* are a central topic in complex analysis. M. C. Escher's *Circle Limit* woodcuts, inspired by a diagram of the mathematician H. S. M. Coxeter, show tilings of the Poincaré disc by congruent fish and angels, which shrink towards the edge because hyperbolic distances blow up there.

Try this next

1. **A numerical check.** Take $z = \frac{1}{2}$ and $w = -\frac{1}{2}$. Compute both sides of the inequality. Then compute the hyperbolic distances $d(z, 0)$, $d(0, w)$ and $d(z, w)$ and check the triangle inequality is an equality here. Why?

2. **The mixed case.** Find z, w with $|z| < 1 < |w|$ for which the inequality fails.

3. **An isometry.** Let $f(z) = \dfrac{z - a}{1 - a^{*}z}$ with $|a| < 1$. Show that $|f(z)| = 1$ whenever $|z| = 1$, and that $f(a) = 0$. Show that $\rho(f(z), f(w)) = \rho(z, w)$.

4. **Infinity is near.** Show that the hyperbolic distance from 0 to r (with $0 < r < 1$) is $\ln \frac{1+r}{1-r}$, and that it tends to infinity as $r \to 1$. How close to the boundary must a point be to be hyperbolic distance 10 from the centre?

Further reading

- **Caroline Series, "Hyperbolic geometry" lecture notes** (University of Warwick, freely available). A clear first course, starting from the Poincaré disc.
- **Jeremy Gray, *Ideas of Space*** (Oxford University Press). The history of non-Euclidean geometry from Euclid to Einstein.
- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). Chapter 6 covers non-Euclidean geometry and Möbius transformations, with beautiful pictures.

The full guided solution is in Section 5.8:
https://beyondstep.keithmcnulty.org/ch05/poincare-disc.html